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A novel lightweight aerodynamic design for the wings of hypersonic vehicles cruising in the upper atmosphere

Zhi Chen [a,c], Fei Huang [a,c], Xuhong Jin [a,b,c,*], Xiaoli Cheng [a,c], Bing Wang [b]

[a] China Academy of Aerospace Aerodynamics, Beijing, 100074, China
[b] School of Aerospace Engineering, Tsinghua University, Beijing, 100084, China
[c] Laboratory of Aero-thermal Protection Technology for Aerospace Vehicles (LAPTAV), China Aerospace Science and Technology Corporation (CASC), Beijing, 100074, China

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ABSTRACT

Base on the concept of molecular mean free paths in high-altitude rarefied aerodynamics, a novel conceptual lightweight design is proposed and applied to the aerodynamic design of wings of hypersonic vehicles cruising in the upper atmosphere between 120 km to 300 km. The lightweight scheme proposed here is promising in hypersonic vehicles cruising in the upper atmosphere in the sense that it is able to not only reduce the weight of a wing by about 14%, but also save the material by the same amount. A comprehensive aerodynamic analysis indicates that the lightweight aerodynamic design can apply to a wide range of cruising altitudes and Mach numbers. In comparison with the conventional flat-plate wing, the lightweight wing always has smaller drags and larger effective lift-to-drag ratios, proving the feasibility of lightweight design for hypersonic vehicles at high altitudes. Moreover, as the cruising altitude is increased, the lightweight wing has even better lift and lift-to-drag-ratio performances, further demonstrating the excellent promise of lightweight wings for hypersonic vehicles cruising in the upper atmosphere. This lightweight design concept is of great value because the region of the upper atmosphere is relatively unexplored, and hypersonic vehicles cruising in this region can be used for high-resolution earth surveillance, accurate gravity/magnetic field mapping, and global ocean climate exploring.

1. Introduction

In aerospace applications, an effective way to increase energy efficiency and reduce fuel consumption is reducing the mass of aircraft, as a lower mass requires less lift force and thrust during flight [1]. For example, a 20% weight reduction can result in 10%–12% fuel efficiency improvement for the Boeing 787. Besides, in astronautic applications, the final speed that a rocket can gain is directly proportional to the ratio of the original mass to the mass after the fuel has been exhausted [2]. So in spacecraft design, this ratio is made as large as possible by reducing the mass of the spacecraft and the rocket, which means that the initial mass of the spacecraft-rocket combination is almost all fuel. It comes to light that reducing fuel consumption and increasing energy efficiency will undoubtedly bring in substantial economic benefits. In addition to energy-efficiency amelioration, flight performance improvements such as larger acceleration, better maneuverability, higher operational flexibility, longer flight range, lower maintainability cost, and more reliable safety performance, could also be achieved by lightweight design [3].

In aerospace applications, the aim of lightweight design is to obtain the same or enhanced aerodynamic or structural performance using less material or materials with lower density. A direct approach to reach this goal is to apply advanced lightweight materials to crucial aerospace components and systems, and the rapid growth in aerospace industry gives an impetus for the fast development of new spacecraft materials [4]. Another effective technique to achieve lightweight design is optimizing the distributions of material on aerospace vehicles to reduce material use and keep the same aerodynamic or structural performance. Conventional optimization methods include size, shape and topology optimization [5]. Although using advanced lightweight materials in spacecraft manufacturing can achieve both weight reduction and performance improvement easily and effectively, lightweight high-performance materials are much more expensive than conventional materials with the same performance [6]. Therefore, the optimization method remains an extensively explored and utilized technique in lightweight design and receives increasing interest in aerospace applications [7,8].

* Corresponding author at: China Academy of Aerospace Aerodynamics, Beijing, 100074, China.
E-mail address: jxh17@mails.tsinghua.edu.cn (X. Jin).

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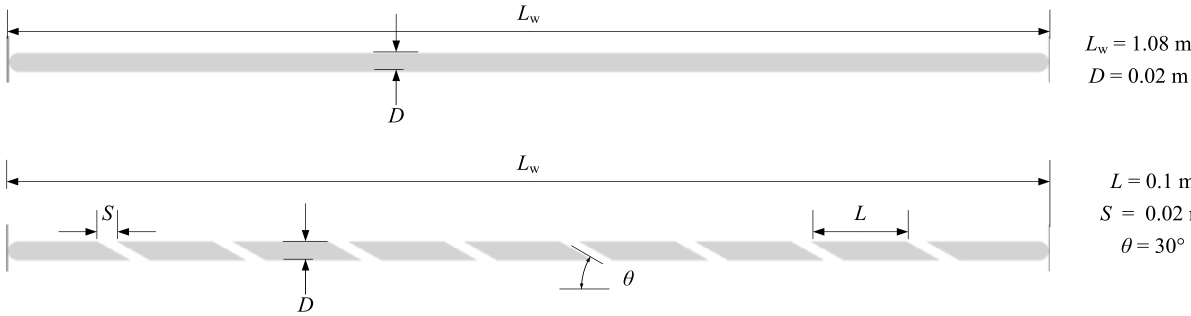


Fig. 1. Schematic drawings of a conventional flat-plate wing and a lightweight wing.

Due to its extreme importance, many examples of lightweight design have been successfully applied to aerospace vehicles from the design phase to manufacture. For example, the Zephyr 7, which is a high-altitude pseudo-satellite solar powered UAV from Airbus, currently holds the world record for the longest absolute flight. This is partly benefitting from increased energy efficiency by lightweight [4]. Rosenblum proposed a concept design of a box wing aircraft where shape optimization is employed in the wing design [4]. Aerodynamic efficiency could be increased by using a box-wing structure, i.e. lower induced drag force result from the box wing compared with conventional wing configurations.

Hypersonic vehicles cruising in the upper atmosphere between 120 km to 300 km are crucial for both national defence and space exploration purposes [9]. Due to the extensive range of speeds the operational envelope covers, one of the greatest concerns with a hypersonic cruising vehicle is the aerodynamic performance and the designs have to fulfil several aerodynamic requirements [10,11]. A particularly important figure of merit in aerodynamics is the ratio of lift to drag, the so-called L/D ratio. Other things being equal, a higher L/D means better performance [12]. For example, in the case of a horizontal take-off vehicle, it is essential to provide large amounts of lift during the take-off and climb phase. Making full use of aerodynamic lift can considerably reduce the thrust required compared to lifting the vehicle vertically, but this mainly depends on the L/D ratio of the vehicle [10,13]. Unfortunately, for hypersonic flight vehicles, the lift-to-drag ratio L/D, as the free-stream Mach number increases, decreases rather dramatically. This undesirable result is brought about by the rapidly growing shock-wave strength as Mach number rises, with consequent large increases in wave drag [14–18]. Therefore, in the process of lightweight design, it is important to obtain the same or larger lift-to-drag ratio L/D.

In the present work, we propose a novel conceptual lightweight design based on the concept of molecular mean free paths in high-altitude rarefied aerodynamics, and apply the lightweight concept to the aerodynamic design of wings of hypersonic vehicles cruising in the upper atmosphere between 120 km to 300 km. The lightweight aerodynamic design is assessed by analysing the intuitive flow-field patterns and integral aerodynamic properties, with a range of flight altitudes and free-stream Mach numbers considered. We note that, to this authors' knowledge, the lightweight design concept proposed in this paper is the first aerodynamic design of wings of hypersonic vehicles cruising in the upper atmosphere. This lightweight design concept shows great promise because the region of the upper atmosphere is relatively unexplored [19], and hypersonic vehicles cruising in this region can be used for high-resolution earth surveillance, accurate gravity/magnetic field mapping, and global ocean climate exploring [20].

2. Lightweight aerodynamic design

2.1. Molecular mean free path

Firstly, let us introduce the concept of the molecular mean free path, denoted by λ, which is defined as the mean distance travelled by a gas molecule between successive collisions with other gas molecules. It can be shown that the mean free path is inversely proportional to the number of molecules per unit volume and inversely proportional to the cross-sectional area of a molecule. According to the kinetic theory of gas, the molecular mean free path is given by [21]

$$\lambda = 1/\left(\sqrt{2}\pi d^2 n\right) \tag{1}$$

where d is the separation of the centres of the two molecules involved, and n stands for the number of molecules per unit volume, i.e. the number density. It is evident that the more molecules there are and the larger the molecule, the shorter the mean distance between collisions.

In the upper atmosphere, as the flight altitude rises from 120 km to 300 km, referring to the U.S. Standard Atmosphere 1976 [22], the free-stream molecular number density n_∞ decreases from 5.10694×10^{17} m^{-3} to 6.50953×10^{14} m^{-3}, and the free-stream molecular mean free path λ_∞ ranges from 3.3 m to 2595.4 m. The extremely large free-stream molecular mean free path successively results in fair large molecular mean free paths in the vicinity of hypersonic vehicles cruising in the upper atmosphere.

2.2. Scheme of lightweight design

Having introduced the concept of molecular mean free path, let us formulate the scheme of lightweight design in detail. Consider the simplified geometric model of a wing of a hypersonic vehicle, i.e. a conventional flat-plate wing with blunt leading/tail edges, as showed in the upper frame of Fig. 1. In order to reduce the mass of the wing, we replace the flat plate with an array of flat plates, and there are several gaps between the plates, as is illustrated in the lower frame of Fig. 1. Note that, because a hypersonic vehicle usually cruising at an angle of attack, each gap is inclined at a certain angle to improve the aerodynamic performance. The neighbouring flat-plate segments can be connected with lightweight and high-strength slender bars. The rationale behind the aforementioned lightweight scheme can be explained on a physical basis, as follows.

In the upper atmosphere, the free-stream molecular mean free path is generally on the same order of or much larger than the characteristic size of a hypersonic cruising vehicle, so we can be sure that the spacing between plates is on the same order of magnitude as or smaller than the local molecular mean free paths. It follows directly from the high-altitude rarefied aerodynamics that the flow-field patterns over the wing can hardly perceive the presence of the gaps. Thus, the pressure and shear stress distributions over the wing surface does not almost influence by the presence of the gaps. Because the only mechanism that nature has for communicating a force to a body moving through a fluid are pressure and shear stress distributions on the body surface [12], the lightweight design scheme of introduction of gaps will have negligible effects on aerodynamic properties, including drag, lift and lift-to-drag ratio, of the wing. This is the essence of the lightweight design in the upper atmosphere.

2.3. Lightweight efficiency

Let us consider lightweight efficiency of the wing of a hypersonic cruising vehicle, which is defined as

$$\eta = m_{\text{saved}}/m_{\text{initial}} \tag{2}$$

where m_{initial} represents the initial mass of the wing, and m_{saved} denotes the saved mass due to the lightweight aerodynamic design.

As a specific example of lightweight aerodynamic design, we consider an original flat-plate wing, which has a length L_{w} of 1.08 m and a thickness D of 0.02 m, as is labelled in Fig. 1. For the lightweight wing, each flat plate in the array has a length L of 0.1 m, except the two at the head and tail, and all plates have the same thickness as the original flat-plate wing. The spacing S between successive flat-plate segments is 0.02 m, and the angle θ at which each gap is inclined has the value of 30°. Substituting the geometric parameters above into Eq. (2) and assuming a uniform distribution of mass over the wing, the lightweight efficiency can be obtained as

$$\begin{aligned}\eta &= \frac{8SD}{(L_{\text{w}}-D)D+\pi(D/2)^2}\\ &= \frac{8\times 0.02\times 0.02}{(1.08-0.02)\times 0.02+0.01^2\pi} = 14.874\%\end{aligned} \tag{3}$$

Therefore, the lightweight scheme is so promising in hypersonic vehicles cruising in the upper atmosphere in the sense that it can not only reduce the weight of a wing by 14.874%, but also save the material used for the wing by the same amount. However, as a class of aerodynamic design, before claiming the usefulness, some aerodynamic performances must be assessed.

3. Computational tools

3.1. Computational method

Due to the extremely low atmospheric density, flows over hypersonic vehicles in the upper atmosphere are characterised by transitional and free molecular flows, both of which fall into the scope of rarefied gas dynamics. Among the few types of computational approaches for complex rarefied gas flows, the most widely used technique is the direct simulation Monte Carlo (DSMC) method, which was pioneered by Bird [23,24]. During the development of more than 55 years, the importance of the DSMC technique is its ability to fill the void in the analysis methodology for transitional flows [25]. The DSMC technique simulates the same physics as the Boltzmann equation without solving partial differential equations. It has demonstrated that DSMC converges to the solution of Boltzmann equation in the limit of an infinitely large number of simulation particles [26].

The DSMC method emulates gas flows as a collection of simulation particles. Each particle, which stands for a fixed number of real gas molecules, has a position, velocity, and internal energies (for diatomic and polyatomic gas molecules). The state of a particle is stored and updated with time as the particle moves, collides with another particle and interacts with the solid surface inside the computational domain [27]. The assumption of dilute gas, that the mean molecular space in the gas is much larger than the mean molecular diameter, allows molecular motions to be decoupled from molecular collisions [24]. Particle movement is calculated deterministically, while collisions are treated with statistically. The simulation is always computed as unsteady flow, and a steady flow solution can be attained as the large-time state of the simulation [28].

The DSMC code used in this work is the free open-source SPARTA (SPARTA stands for Stochastic PArallel Rarefied-gas Time-accurate Analyzer) [29], which is under active development at Sandia National Laboratory. SPARTA overlays the simulation domain with a hierarchical Cartesian grid to track particles, perform collisions, and sample requested macroscopic quantities, and can run efficiently on massive parallel computing architectures. In the present work, internal energy exchange is simulated using the Larsen-Borgnakke scheme [30], the molecular collision is modelled using the variable hard sphere (VHS) model [31] and the no time counter collision sampling technique [32].

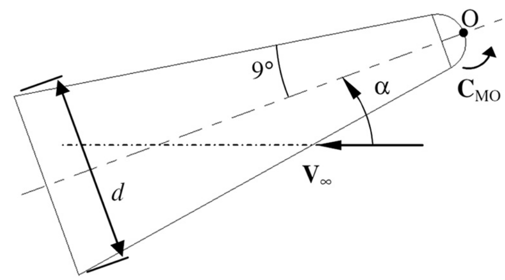


Fig. 2. Schematic drawing of the blunted-cone geometry.

Table 1
Free-stream conditions for the blunted-cone case.

Gas	V_∞ (m/s)	ρ_∞ (kg/m^3)	T_∞ (K)	Ma_∞
N_2	2478	6.154×10^{-5}	143.5	10.15

3.2. Validation case

As an open-source code that is widely used in the rarefied gas dynamics community, the SPARTA code has been applied to many typical problems in the transitional and free molecular regime, in which the validity has been confirmed [33,34]. In order to check its accuracy in high-altitude hypersonic flows quantitatively, we apply the SPARTA code to the flow over a blunted cone, which has a nose radius of 2.286 mm, a semi-cone angle of 9° and a base diameter of 15.24 mm, as is illustrated in Fig. 2. The blunted-cone problem is chosen here because experimental [35] and computational aerodynamic data [36] are available. The free-stream gas is exclusively composed of molecular nitrogen (N_2), and other free-stream parameters, including free-stream speed V_∞, mass density ρ_∞, temperature T_∞ and Mach number Ma_∞, are tabulated in Table 1. Note that the blunted-cone surface temperature is kept constant at 600 K and six angels of attack $\alpha = 0°$, 5°, 10°, 15°, 20° and 25°, are considered. The pitching moment for the 9° blunted-cone is taken with respect to the apex O in Fig. 2. It should be noted that the computational grid at each angle of attack is composed of non-uniform cells in the sense that cell clustering is employed in the compression region. As an example, the computational grid in the case of $\alpha = 25°$ is made up of 543,604 cells, and the grid on the section defined as $z = 0$ is schematically illustrated in Fig. 3.

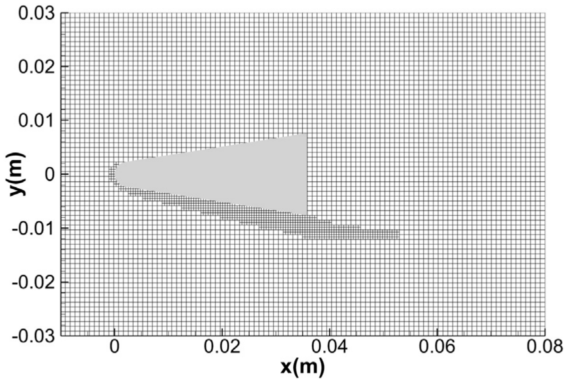


Fig. 3. Computational grid on the section of $z = 0$ at $\alpha = 25°$.

The flow-field pressure contours and streamline patterns on the longitudinal symmetric plane (the plane defined as $z = 0$) for $\alpha = 0°$ and 25° are plotted in Fig. 4. As is expected, the flow field is symmetric about the plane $y = 0$ at $\alpha = 0°$. Note that, even at as large as an angle of 25°, no flow separation phenomenon is observed due to the strong viscous effect (low-Reynolds effects) involved in the rarefied gas flows. In Fig. 5, the lift-to-drag ratios and pitching moment coefficients obtained here are compared with the computational results given by the MONACO code [36] and the experimental results of AEDC-VKF Wind Tunnel [35]. In order to take the uncertainty of gas-surface interaction (GSI) into account, two values of Maxwell accommodation coefficient ($\sigma = 1$ and 0.85) are considered, which is physically defined as the fraction of diffuse reflections, are considered. $\sigma = 1$ means that all GSIs are completely diffuse reflections, while $\sigma = 0$ indicates that all GSIs are purely specular reflections. As shown in Fig. 5, both the lift-to-drag ratios and the pitching moment coefficients obtained here agree reasonably well with MONACO results and wind-tunnel data, confirming the reasonable accuracy of the SPARTA code in simulating rarefied hypersonic flows.

4. Free-stream and computational conditions

Consider a hypersonic vehicle cruising at three altitudes ($H =$ 80 km, 100 km and 120 km), and the atmosphere is comprised of 76.3% molecular nitrogen (N_2) and 23.7% molecular oxygen (O_2) in volume. The three altitudes are selected such that they represent the top of near space (20 km-100 km) and the bottom of upper atmosphere (120 km-300 km). The corresponding free-stream parameters are listed in Table 2, which can be retrieved from the U.S. Standard Atmosphere 1976 [22]. Note that, in order to assess the applications of the lightweight design to a large range of hypersonic cruising vehicles, three free-stream Mach numbers ($Ma_\infty = 5$, 10 and 25) and eight angles of attack ($\alpha = 0°$, 5°, 10°, 15°, 20°, 25°, 30° and 45°) are taken into account. The surface is assumed completely diffuse and its temperature T_w is kept constant at 300 K.

5. Verification of grid resolution

For an accurate DSMC simulation, it is well known that there are inherent constraints that need to be conformed to with re-

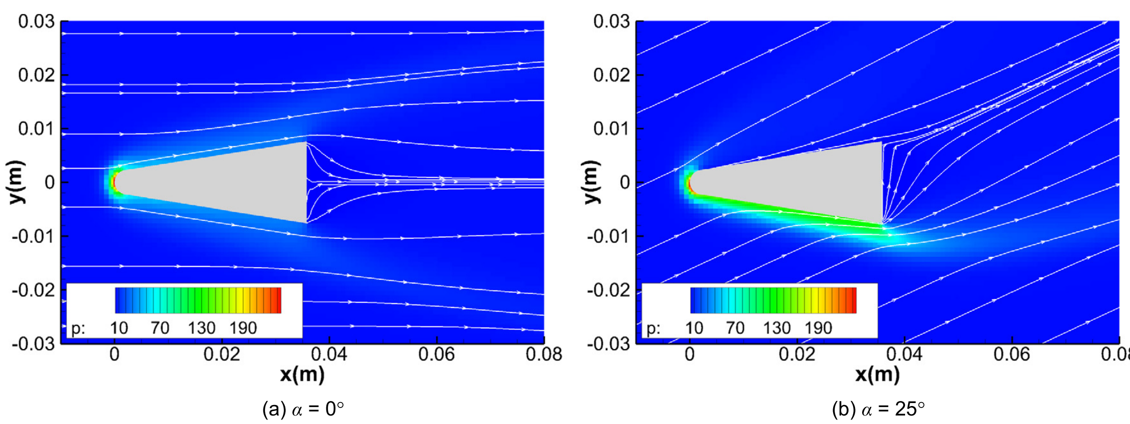


Fig. 4. Pressure contours and streamline patterns for the blunted-cone case (Unit: Pa). (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

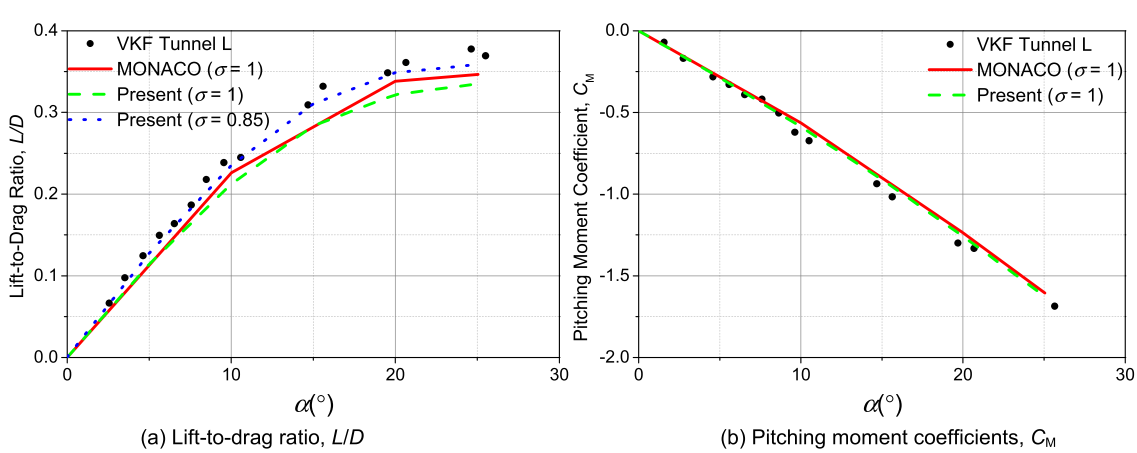


Fig. 5. Aerodynamic properties for the blunted-cone case.

Table 2
Free-stream conditions for the current study.

Altitude (H)	80 km	100 km	120 km
Mach number (Ma_∞)	5, 10, 25	5, 10, 25	5, 10, 25
Angle of attack (α)	0°, 5°, 10°, 15°, 20°, 25°, 30°, 45°	0°, 5°, 10°, 15°, 20°, 25°, 30°, 45°	0°, 5°, 10°, 15°, 20°, 25°, 30°, 45°
Temperature (T_∞)	198.62 K	195.08 K	360.00 K
Pressure (p_∞)	1.05 Pa	3.20×10^{-2} Pa	2.54×10^{-3} Pa
Mass density (ρ_∞)	1.844×10^{-5} kg/m^3	5.604×10^{-7} kg/m^3	2.222×10^{-8} kg/m^3
Mean free path (λ_∞)	4.402×10^{-3} m	1.421×10^{-1} m	3.308 m

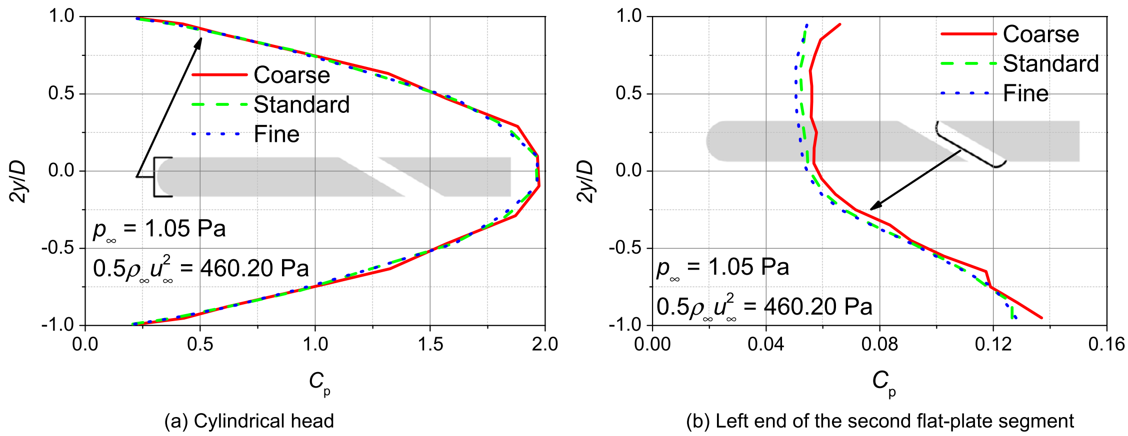


Fig. 6. Distributions of pressure coefficients over some parts for three sets of grids at $H = 80$ km, $Ma_\infty = 25$ and $\alpha = 0°$.

gards to the cell size. Thus, in this section, a grid-independence study was performed using the lightweight wing shown in Fig. 1. Note that in order to verify the grid resolution, the cruising altitude was fixed at 80 km, and the free-stream Mach number was kept at 25 and the angle of attack was fixed at 0°. Note that three different sets of Cartesian grids (coarse, standard, and fine) were used, and each set of grid is made up of non-uniform cells in the sense that cell clustering is utilised in the regions close to the lightweight wing. In this way, the coarse, standard, and fine grids are composed of 154,470, 346,815, and 611,936 cells, respectively.

Distributions of pressure coefficients over the cylindrical head and the left end of the second plate segment are plotted in Fig. 6 for the three sets of grids mentioned above. Note that pressure coefficients are defined as

$$C_p = (p - p_\infty)/\left(0.5\rho_\infty u_\infty^2\right) \tag{4}$$

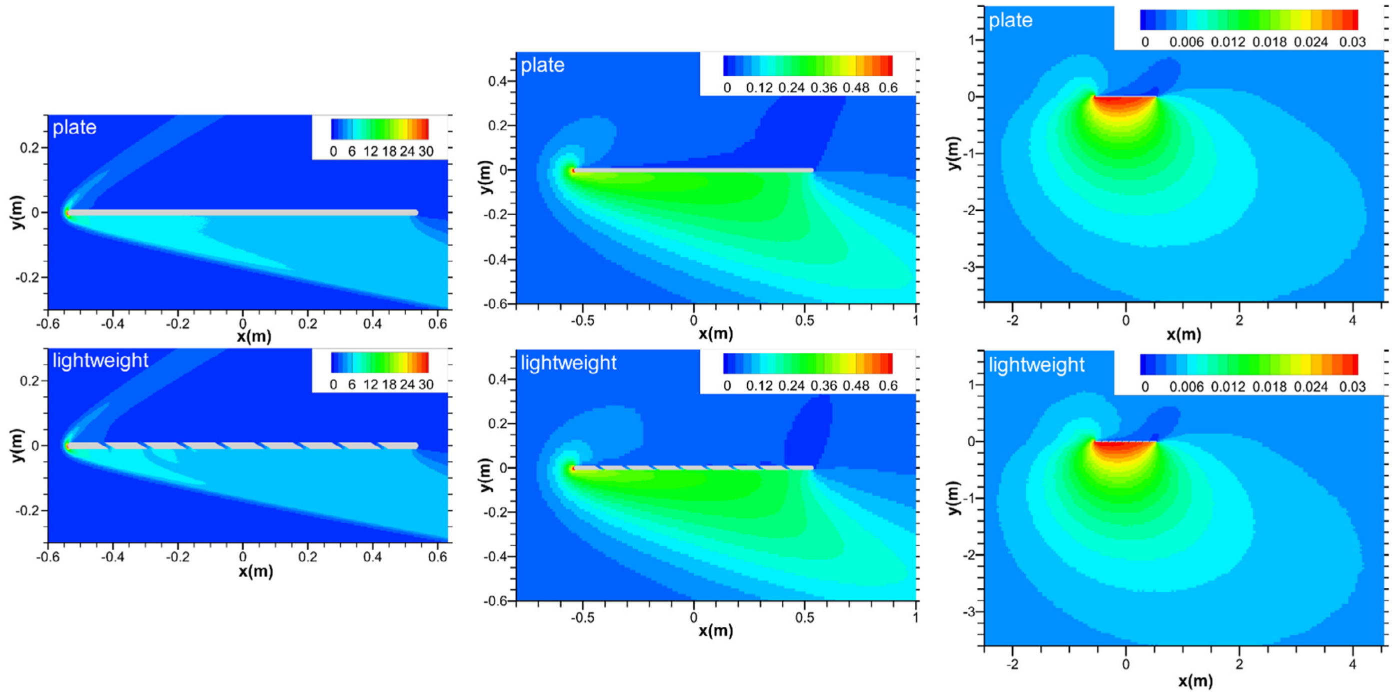
Note that in Fig. 6, the abscissa axis stands for the pressure coefficient, while the ordinate axis represents the vertical displace normalised by the half of the thickness of the lightweight wing, namely $y/(D/2)$. It is evident, according to these graphs, pressure coefficients on the cylindrical head are rather insensitive to the three sets of grids considered here, and there is negligible discrepancy in pressure coefficient on the left end of the second plate segment between the standard and fine grids. Therefore, the standard grid consisting of 346,815 cells is deemed to be enough for the computation of the flow over the lightweight wing considered here, and we can conclude the computational results using the standard grid is independent of grid.

6. Flow-field patterns

Before embark on the quantitative assessment of the aerodynamic properties of the flat-plate and lightweight wings, it is instructive to examine the flow-field characteristics graphically. Thus, pressure contours for the flat-plate and lightweight wings at the three altitudes ($H = 80$ km, 100 km and 120 km) are presented in Fig. 7. For the sake of brevity, only the contours for the case of $Ma_\infty = 5$ and $\alpha = 15°$ are given, but it is adequate to give us a clear understanding of the flow-field characteristics.

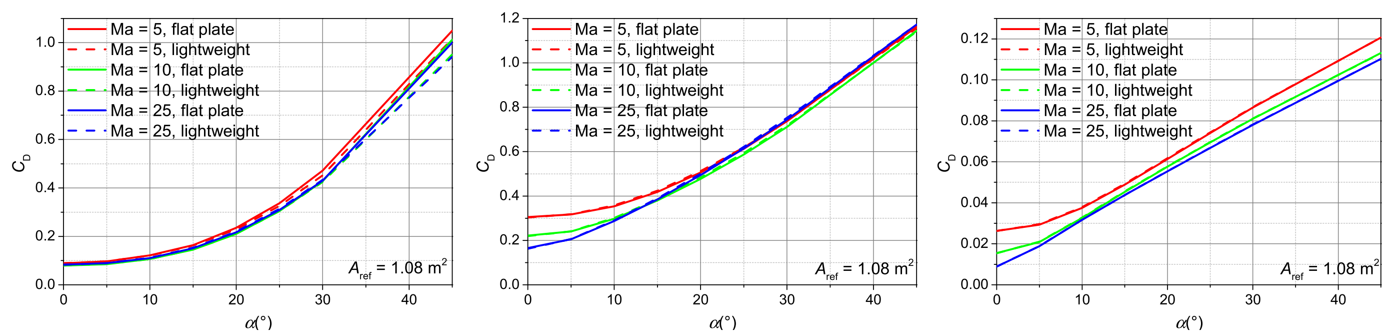
At an altitude of 80 km, the free-stream molecular mean free path is 4.402 mm, and molecular mean free paths in the compression region usually have an even smaller value. Therefore, mean free paths just below the wing are much smaller than the spacing S between successive flat plates, causing that a sizeable fraction of gas can flow through the gaps from the comparison region at the bottom of the wing upward to the expansion region at the top of the wing. As is shown in Fig. 7a, the gas leak through the gaps directly results in a so-called “pressure leak”, which, on the one hand, helps to lessen the pressure just below the wing; on the other hand, contributes to the increase in the pressure just above the wing. Besides, the gas leak makes the oblique shock wave below the lightweight wing stand slightly closer to the wing in comparison the flat-plate case.

As the cruising altitude rises to 100 km, the free-stream mean free path becomes 0.1421 m, and this value is much larger than the spacing S. Thus, mean free paths just below the wing are on the same order of magnitude as the spacing S, and only a small fraction of gas molecules can travel through the gaps. In other words, increasing the cruising altitude can weaken the gas and pressure leaks; thus, the pressure contour of the lightweight wing is almost the same as that of the flat-plate wing, except in the leeward region located at the head and tail of the wing. Moreover, when the cruising altitude further grows to 120 km, mean free paths just below the wing are primarily much larger than the spacing S, and few gas molecules can move through the gaps. That is to say, the gas and pressure leaks are negligibly weak, and the lightweight design has a paltry effect on the flow-field patterns. Therefore, the pressure contours of the flat-plate and lightweight wings are the essentially same. The above comparison of flow-field pressure contours provides graphical evidence of the feasibility of lightweight wing in the upper atmosphere.



(a) H = 80 km (b) H = 100 km (c) H = 120 km

Fig. 7. Flow-field pressure contours for flat-plate and lightweight wings at $Ma_\infty = 5$ and $\alpha = 15°$. (Unit: Pa.)



(a) H = 80 km (b) H = 100 km (c) H = 120 km

Fig. 8. Drag coefficients of the flat-plate and lightweight wings at the three altitudes.

7. Aerodynamic properties

7.1. Drag properties

Let us now compare the aerodynamic performance of the lightweight wing with that of the conventional flat-plate one. First, consider the variations of drag properties with angles of attack for a range of free-stream Mach numbers and cruising altitudes, as plotted in Fig. 8. The drag coefficient is defined as

$$C_D = D/\left(0.5\rho_\infty u_\infty^2 A_{ref}\right) \tag{5}$$

where the reference area A_{ref} is the platform area per unit span length, namely $A_{ref} = L_w$ m^2.

At an altitude of 80 km, resulting from the perceptible pressure leak present in the lightweight wing, the drags due to pressure on the lower surface of the wing decrease slightly, producing a smaller drag for the lightweight wing in comparison with the conventional flat-plate one. However, as the cruising altitude rises to 100 km and 120 km, drag coefficients of lightweight and flat-plate wings are virtually the same, indicating that the effects of gas and pressure leaks are weakened so substantially that they have negligible influence on drag properties. Also, note the minor discrepancy in drag coefficients among the three free-stream Mach numbers, especially for all the angles considered here at the altitude of 80 km and for moderate and large angles at the altitudes of 100 km and 120 km. This is the manifestation of Mach-number independence principle in hypersonic aerodynamics, which states that, at high enough Mach numbers, certain aerodynamic quantities, such as pressure coefficients and pressure-drag coefficients, become essentially independent of Mach number [14].

7.2. Lift properties

Lift properties play a strong role in the preliminary design and performance analysis of hypersonic cruising vehicles. Thus, we proceed with the variations of lift properties with angles of attack, as shown in Fig. 9. The lift coefficient is defined as

$$C_L = L/\left(0.5\rho_\infty u_\infty^2 A_{ref}\right) \tag{6}$$

It is evident that lightweight wings' lift coefficients are generally lower than flat-plate wings', showing that lifts are more sensitive to gas and pressure leaks. This is because that pressure is

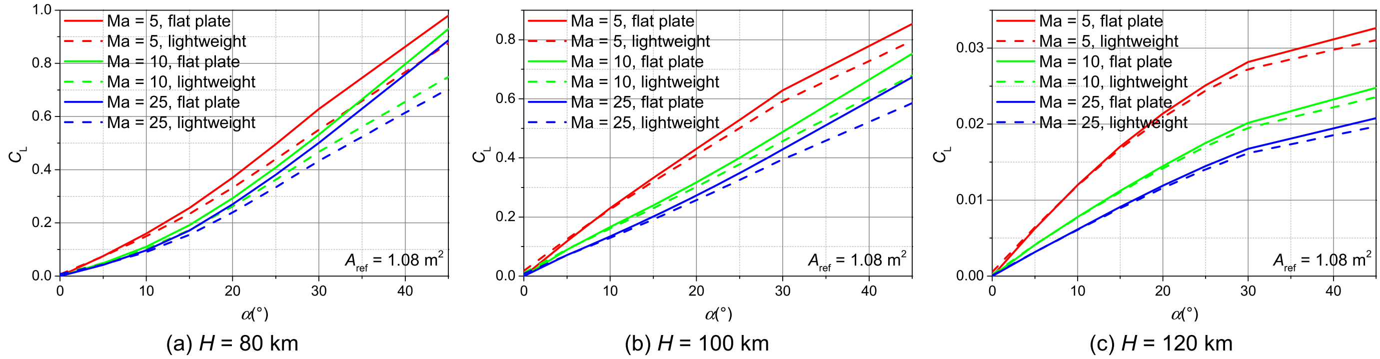


Fig. 9. Lift coefficients of the flat-plate and lightweight wings at the three altitudes.

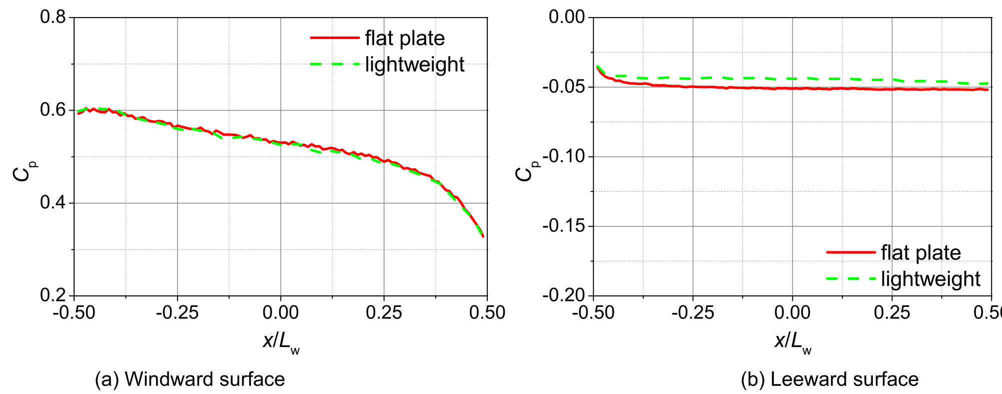


Fig. 10. Distributions of pressure coefficients over the surfaces of flat-plate and lightweight wings at $H = 100$ km, $Ma_\infty = 5$ and $\alpha = 20°$.

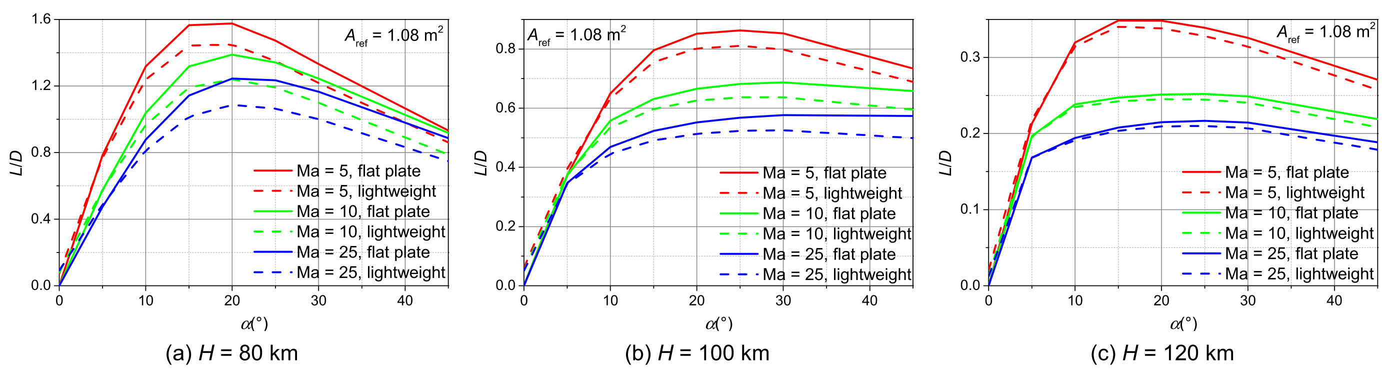


Fig. 11. Lift-to-drag ratios of the flat-plate and lightweight wings at the three altitudes.

primarily responsible for the generation of lift, and it is more easily influenced by the presence of gas and pressure leaks. This can be seen in detail from Fig. 10, which is the distributions of pressure coefficients over the surfaces of flat-plate and lightweight wings at $H = 100$ km, $Ma_\infty = 5$ and $\alpha = 20°$. According to the graphs in Fig. 10, we can conclude that gas and pressure leaks present in the lightweight wing have negligible effects on the pressure on the windward surface, while bring about an increase in the pressure on the leeward, consequently leading to a slight decrease in the lift coefficient.

However, referring to Fig. 9 again, it is well worthwhile to note that, as the cruising altitude rises, the two kinds of wings' lift performances become asymptotically closer, indicating that lightweight wings will achieve better lift performances at higher altitudes in the upper atmosphere. The extremely low lift coefficients in Fig. 9 should also be noted. These are typical lift of hypersonic vehicles at high altitudes. Moreover, lift performances get even worse when these vehicles cruise at even larger Mach numbers or higher altitudes. This phenomenon emphasizes the special importance of remaining the same or better lift performance in the lightweight design of hypersonic vehicles cruising at high altitudes.

7.3. Lift-to-drag ratios

A true measure of the aerodynamic efficiency of a vehicle shape is its lift-to-drag ratio, so in this subsection, we make a detailed comparison of lift-to-drag ratios between conventional flat-plate and lightweight wings, and the variations of lift-to-drag ratio with angles of attack are presented in Fig. 11.

As far as the difference between flat-plate and lightweight wings is concerned, lift-to-drag ratios follow an analogous behaviour to lift coefficients graphed in Fig. 9 in the sense that lightweight wings have a slightly lower lift-to-drag ratio compared with flat-plate wings. Moreover, the discrepancy in the lift-to-drag ratio between the two types of wings gets increasingly insignifi-

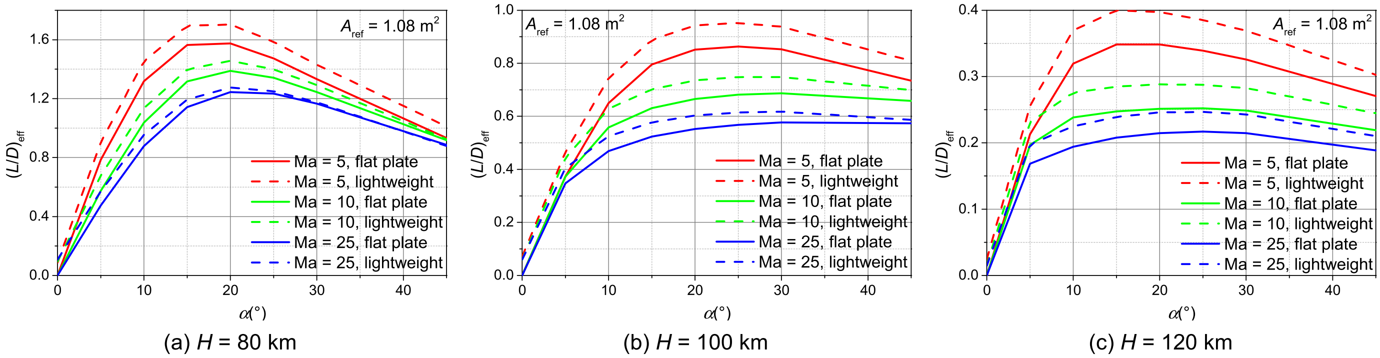


Fig. 12. Effective lift-to-drag ratios of the flat-plate and lightweight wings at the three altitudes.

cant when the cruising altitude grows, which proves the technical feasibility of lightweight design in upper atmosphere from the aerodynamic point of view. It should also be meaningful to notice the exceedingly poor lift-to-drag ratios at large Mach numbers and high altitudes, which is typical characteristics of high-altitude hypersonic aerodynamics. This strongly highlights the fact that it is absolutely essential to achieve the same or better lift-to-drag-ratio performance in the lightweight design of hypersonic vehicles cruising at high altitudes.

7.4. Effective lift-to-drag ratios

In order to quantitatively assess the aerodynamic performance of lightweight wings, we define an effective lift-to-drag ratio $(L/D)_{\text{eff}}$ in the following form:

$$(L/D)_{\text{eff}} = (L/D)/(1-\eta) \tag{7}$$

In the definition above, L/D stands for the conventional lift-to-drag ratio, and η is the lightweight efficiency defined in Eq. (2). Apparently, the effective lift-to-drag ratio represents the lift-to-drag ratio when the lightweight wing use the same quantity of material as the conventional flat-plate wing. Therefore, in essence, it is a true measure of aerodynamic efficiency per unit material. Other things being equal, a higher $(L/D)_{\text{eff}}$ means better aerodynamic performance when the same quantity of material is used.

The variations of effective lift-to-drag ratios with angles of attack are plotted in Fig. 12. For free-stream Mach numbers ranging from 5 to 25 and at altitudes from 80 km to 120 km, the effective lift-to-drag ratios of lightweight wing are always larger than those of flat-plate wing. Moreover, as the cruising altitude is increased, the lightweight wing has an even higher effective lift-to-drag ratio in comparison with the flat-plate wing, further demonstrating the excellent promise of lightweight wings in the upper atmosphere.

8. Three-dimensional lightweight design

The lightweight design discussed above is two-dimensional (2-D), which serves to illustrate the physical aspects behind the lightweight design due to its simplicity. However, in actual engineering, all wings are three-dimensional (3-D), so a 3-D lightweight design must be provided, with its lightweight efficiency assessed.

8.1. Lightweight design and lightweight efficiency

The 3-D lightweight wing is schematically shown in Fig. 13, which is actually the extension of the 2-D lightweight wing in the spanwise direction (z-axis direction). As a specific example of a lightweight design, we consider the 3-D lightweight wing that has a width of 0.5 m. In addition, other geometrical parameters the same as the corresponding 2-D wing and there are two lightweight and high-strength slender bars between neighbouring flat-plate segments (for schematic clarity, these bars are not plotted in Fig. 13).

Assume that the slender bars have diameters of D and have the same mass density as the wing, then according to the definition of lightweight efficiency in Eq. (2), the lightweight efficiency can be obtained as

$$\begin{aligned}\eta &= \frac{8[SDW - 2S\pi(D/2)^2]}{(L_{\text{w}} - D)DW + \pi(D/2)^2 W} \\ &= \frac{8 \times [0.02 \times 0.02 \times 0.5 - 2 \times 0.02 \times 0.01^2\pi]}{(1.08 - 0.02) \times 0.02 \times 0.5 + 0.01^2\pi \times 0.5} = 13.939\%\end{aligned} \tag{8}$$

Note that the lightweight efficiency with the slender bars taken into account is just slightly lower than the lightweight efficiency 14.874% expressed in (3), which does not consider the slender bars. This indicates that the slender bars have negligible influence on lightweight efficiency if they are made from the same material as the lightweight wing.

8.2. Flow-field patterns

In this subsection, we analyse the intuitive flow-field patterns and present flow-field pressure contours on the longitudinal symmetric plane (the plane defined as $z = 0$) for the 3-D flat-plate and lightweight wings at $H = 100$ km, $Ma_\infty = 25$ and $\alpha = 15°$ in Fig. 14. For the purpose of comparison, the corresponding flow-field pressure contours of 2-D flat-plate and lightweight wings are also plotted in this figure.

It is evident that, as in the 2-D case, there is no essential difference in the pressure contours between the flat-plate and lightweight wings, which graphically proves the feasibility of 3-D lightweight wing in the upper atmosphere. It should be noted that, in comparison with the 2-D flat-plate or lightweight wing, the shock wave stands closer to body in the 3-D case. This is because the flow over a 3-D wing has an extra dimension to move out of the way of the wing; the flow can move sideways as well as up and down. In contrast, the flow over a 2-D wing is more constrained; it can only move up and down. This is an example of the three-dimensional relieving effect, which is a general phenomenon for all types of 3-D flows.

8.3. Aerodynamic properties

Having a clear qualitative picture of the flow-field patterns of 3-D flows over flat-plate and lightweight wings, we now turn our attention to some quantitative aerodynamic properties. Variations

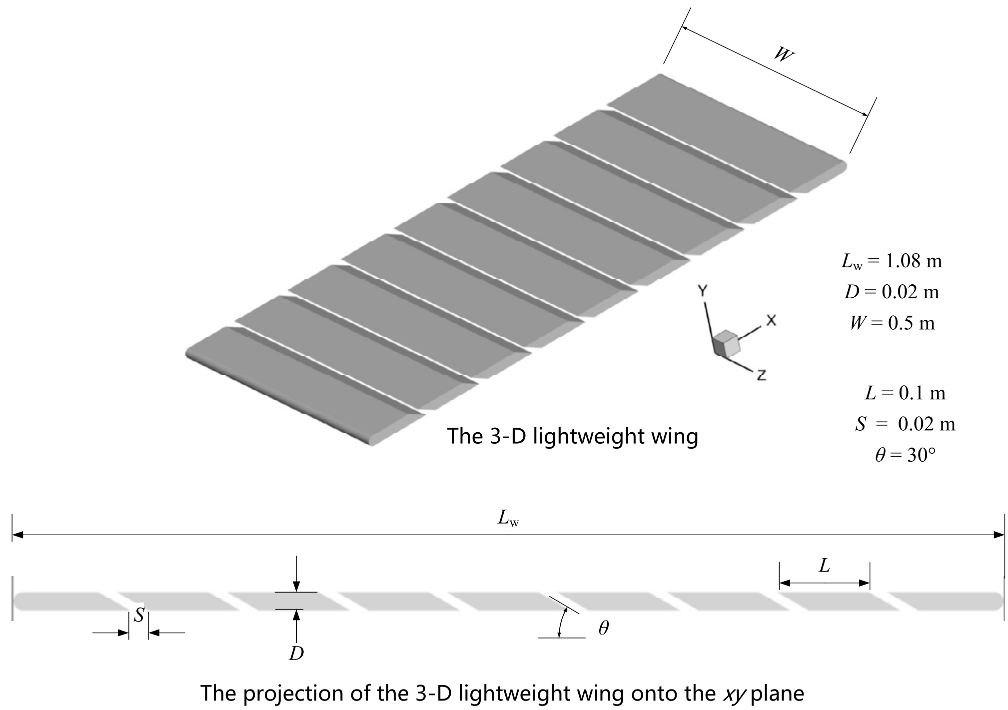


Fig. 13. Schematic drawing of a 3-D lightweight wing.

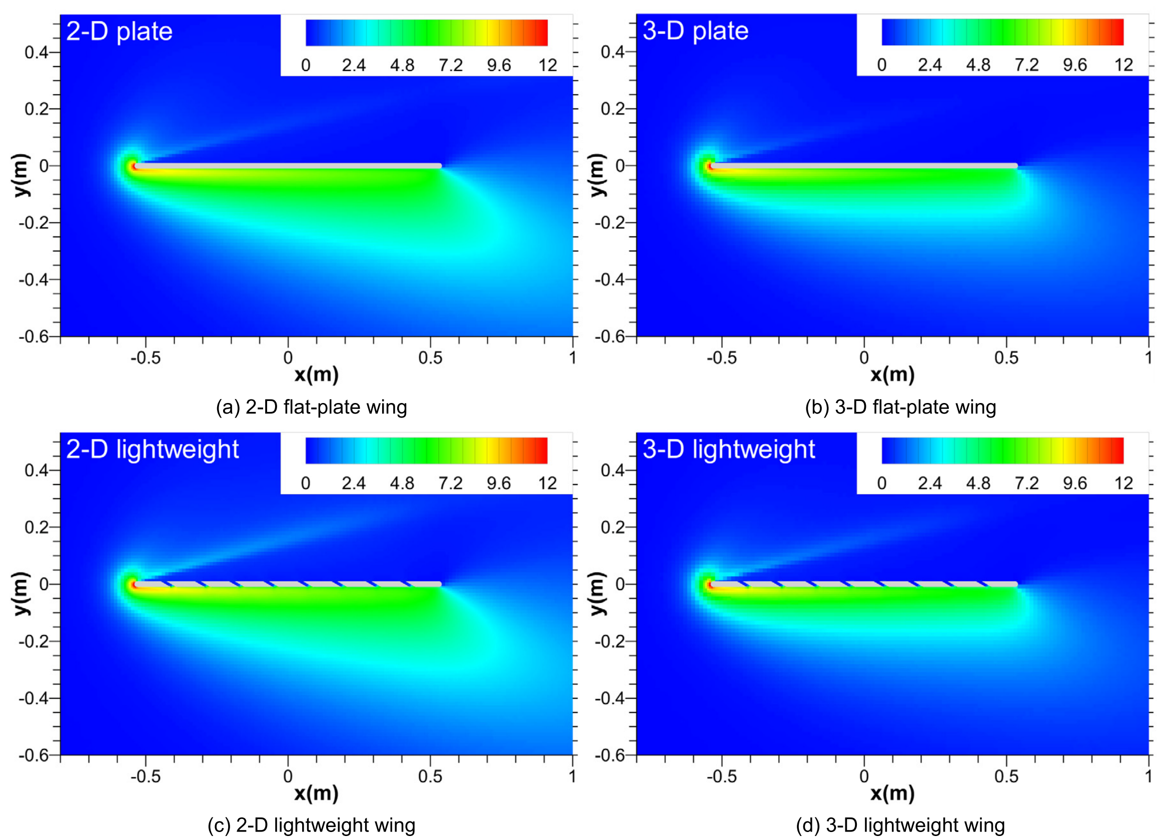


Fig. 14. Flow-field pressure contours for 2-D/3-D flat-plate and lightweight wings at $H = 100$ km, $Ma_\infty = 25$ and $\alpha = 15°$. (Unit: Pa.)

of aerodynamic coefficients with angles of attack for the 3-D flat-plate and lightweight wings at $H = 100$ km and $Ma_\infty = 25$ are shown in Fig. 15. For the purpose of comparison, the corresponding 2-D results are also presented. Note that the reference area A_{ref} for the 3-D flat-plate and lightweight wings is the platform area, namely $A_{\mathrm{ref}} = L_{\mathrm{w}} \cdot W$ m^2.

Analogous with the 2-D case, although the 3-D lightweight wing has a slightly lower lift and lift-to-drag ratio in comparison

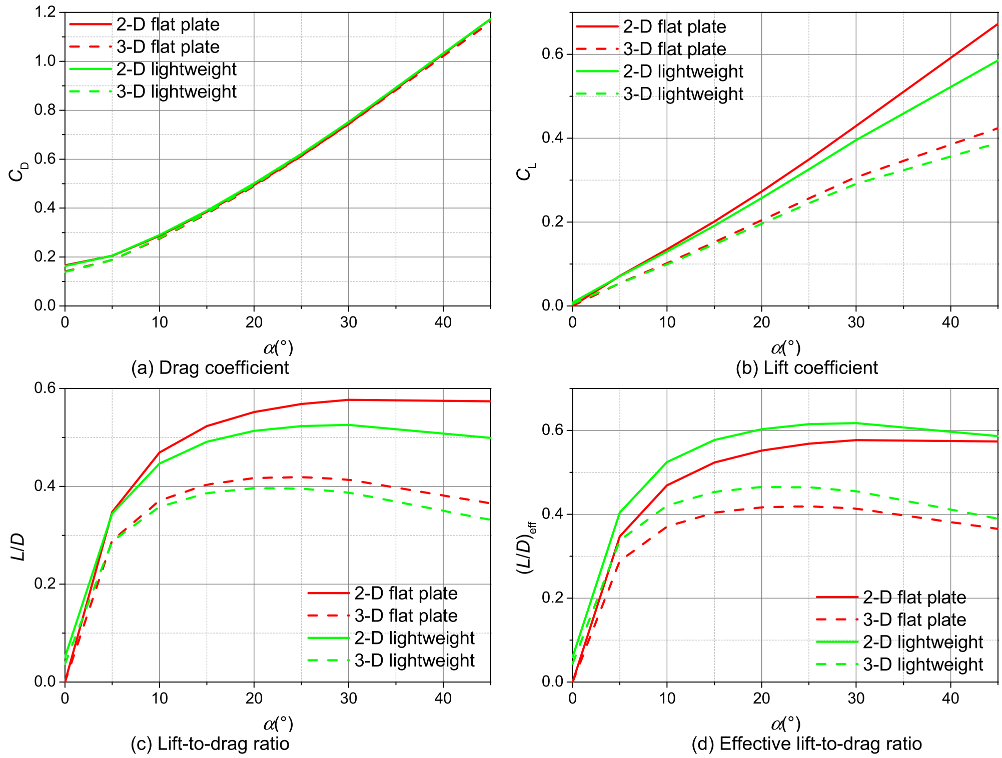


Fig. 15. Aerodynamic properties of 2-D/3-D flat-plate and lightweight wings at $H = 100$ km and $Ma_\infty = 25$.

with the flat-plate wing, it has a higher effective lift-to-drag ratio, which is a true measure of aerodynamic efficiency per unit material, indicating the better aerodynamic performance of the 3-D lightweight design. It is interesting to note that, in comparison with the 2-D flat-plate or lightweight wing, the 3-D wing has a considerable lower lift or lift-to-drag ratio, which is again an example of the three-dimensional relieving effect. That is to say, the spanwise movement relieves the constraint on the flow; the flow does not have to speed up so much to get out of the way of the 3-D wing, and therefore the pressure increment on the wing surface is smaller than the corresponding quantity in 2-D flows. Because pressure on the wing contributes dominantly to the generation of lift, the lift and lift-to-drag ratio of 3-D wing are lower than the corresponding 2-D values.

9. Conclusions

Based on the concept of molecular mean free path in high-altitude rarefied aerodynamics, a novel conceptual lightweight design is proposed and applied to the aerodynamic design of wings of hypersonic vehicles cruising in the upper atmosphere between 120 km to 300 km. A comprehensive aerodynamic analysis has been conducted to assess the lightweight aerodynamic design, with a wide range of cruising altitudes and Mach numbers considered. We now come to several important conclusions.

The lightweight scheme proposed here is promising in hypersonic vehicles cruising in the upper atmosphere in the sense that it is able to not only reduce the weight of a wing by about 14%, but also save the material used for the wing by the same amount. Thus, it is also economically significant because the material used to manufacture wings is usually expensive.

At an altitude of 80 km, free-stream molecular mean paths are quite large, gas and pressure leak phenomena due to the presence of gaps in the lightweight design are noticeable, causing that there is slight difference in flow-field pressure contours between lightweight and flat-plate wings. However, as cruising altitudes rise, gas and pressure leaks become weakened. When the cruising altitude increases to 120 km, gas and pressure leaks so weak that they have negligible effects on the flow-field patterns. The phenomenon provides graphical evidence of the feasibility of lightweight wing in the upper atmosphere.

Resulting from the pressure leak, the lightweight wing has a smaller drag than the conventional flat-plate one, and this discrepancy becomes asymptotically small as cruising altitudes grow. This indicates that the lightweight wing has a better drag performance in comparison with the flat-plate wing, proving the feasibility of lightweight design in the upper atmosphere quantitatively.

Due to the presence of pressure leaks, although the lightweight wing has a slightly lower lift and lift-to-drag ratio in comparison with the flat-plate wing, the two kinds of wings' lift and lift-to-drag-ratio performances become asymptotically closer as the cruising altitude rises. Moreover, in comparison with the conventional flat-plate wing, the lightweight wing always has larger effective lift-to-drag ratios, which is a true measure of aerodynamic efficiency per unit material, further demonstrating the excellent promise of lightweight wings in the upper atmosphere.

However, it is important to note that the lightweight design proposed here is just a conceptual design, and we have to go a long way before we can apply it to engineering applications. For example, further research into the overall aerodynamic properties of a complete vehicle using the lightweight design is suggested to fully assess the aerodynamic performances. In addition, a comprehensive and detailed assessment of strength of structure is considered as important areas.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] E. Ordoukhanian, A.M. Madni, Blended wing body architecting and design: current status and future prospects, Proc. Comput. Sci. 28 (2014) 619–625.
[2] H.D. Young, R.A. Freedman, Sears and Zemansky's University Physics with Modern Physics (13Ed), Addison-Wesley, San Francisco, CA, USA, 2012.
[3] X. Zhu, Z. Guo, Z. Hou, Solar-powered airplanes: a historical perspective and future challenges, Prog. Aerosp. Sci. 71 (2014) 36–53.
[4] L. Zhu, N. Li, P.R.N. Childs, Light-weighting in aerospace component and system design, Propuls. Power Res. 7 (2018) 103–119.
[5] X. Zhang, Y. Chen, J. Hu, Recent advances in the development of aerospace materials, Prog. Aerosp. Sci. 97 (2018) 22–34.
[6] F.C. Campbell Jr, Manufacturing Technology for Aerospace Structural Materials, Elsevier, Oxford, UK, 2011.
[7] T. Ide, M. Otomori, J.P. Leiva, et al., Structural optimization methods and techniques to design light and efficient automatic transmission of vehicles with low radiated noise, Struct. Multidiscip. Optim. 50 (2014) 1137–1150.
[8] O. Sigmund, K. Maute, Topology optimization approaches, Struct. Multidiscip. Optim. 48 (2013) 1031–1055.
[9] S. Gu, H. Olivier, Capabilities and limitations of existing hypersonic facilities, Prog. Aerosp. Sci. 113 (2020) 100607.
[10] D. Sziroczak, H. Smith, A review of design issues specific to hypersonic flight vehicles, Prog. Aerosp. Sci. 84 (2016) 1–28.
[11] D. Expósito, Z.A. Rana, Computational investigations into heat transfer over a double wedge in hypersonic flows, Aerosp. Sci. Technol. 92 (2019) 839–846.
[12] J.D. Anderson, Fundamentals of Aerodynamics (5Ed), McGraw-Hill, New York, NY, USA, 2011.
[13] Z. Wu, Z. Gao, C. Jiang, C. Lee, An in-depth numerical investigation of a supersonic cavity-ramp flow with DDES method, Aerosp. Sci. Technol. 89 (2019) 253–263.
[14] J.D. Anderson, Hypersonic and High-Temperature Gas Dynamics (2Ed), AIAA, Reston, Virginia, USA, 2006.
[15] X.H. Jin, B. Wang, X.L. Cheng, et al., The effects of Maxwellian accommodation coefficient and free-stream Knudsen number on rarefied hypersonic cavity flows, Aerosp. Sci. Technol. 97 (2020) 105577.
[16] Z. Ren, B. Wang, B. Hu, et al., Numerical analysis of supersonic flows over an aft-ramped open-mode cavity, Aerosp. Sci. Technol. 78 (2018) 427–437.
[17] A. Oamjee, R. Sadanandan, Fuel injection location studies on pylon-cavity aided jet in supersonic crossflow, Aerosp. Sci. Technol. 92 (2019) 869–880.
[18] R.C. Palharini, W.F.N. Santon, The impact of the length-to-depth ratio on aerodynamic surface quantities of a rarefied hypersonic cavity flow, Aerosp. Sci. Technol. 88 (2019) 110–125.
[19] S.W. Jackson, R. Marshall, Conceptual design of an air-breathing electric thruster for CubeSat applications, J. Spacecr. Rockets 55 (2018) 632–639.
[20] Y. Li, X. Chen, D. Li, Y. Xiao, P. Dai, C. Gong, Design and analysis of vacuum air-intake device used in air-breathing electric propulsion, Vacuum 120 (2015) 89–95.
[21] G.A. Bird, Molecular Gas Dynamics and the Direct Simulation of Gas Flows, Oxford University Press, Oxford, England, UK, 1994.
[22] COESA, U.S. Standard Atmosphere 1976, U.S. Government Printing Office, Washington, DC, 1976.
[23] G.A. Bird, Approach to translational equilibrium in a rigid sphere gas, Phys. Fluids 6 (1963) 1518–1519.
[24] G.A. Bird, Monte Carlo simulation of gas flows, Annu. Rev. Fluid Mech. 10 (1978) 11–31.
[25] I.D. Boyd, Computation of hypersonic flows using the direct simulation Monte Carlo method, J. Spacecr. Rockets 52 (2015) 38–53.
[26] W. Wagner, A convergence proof for Bird's direct simulation Monte Carlo, J. Stat. Phys. 66 (1992) 1011–1044.
[27] M.A. Gallis, Maximum entropy analysis of chemical reaction energy dependence, J. Thermophys. Heat Transf. 10 (1996) 217–223.
[28] M.A. Gallis, R.B. Bond, J. Torczynski, A kinetic-theory approach for computing chemical-reaction rates in upper-atmosphere hypersonic flows, J. Chem. Phys. 138 (2009) 124311.
[29] S.J. Plimpton, S.G. Moore, A. Borner, A.K. Stagg, T.P. Koehler, J.R. Torczynski, M.A. Gallis, Direct simulation Monte Carlo on petaflop supercomputers and beyond, Phys. Fluids 31 (2019) 086101.
[30] C. Borgnakke, P.S. Larsen, Statistical collision model for Monte Carlo simulation of polyatomic gas mixture, J. Comput. Phys. 18 (4) (1975) 405–420.
[31] G.A. Bird, Monte Carlo simulation in an engineering context, in: Sam S. Fisher (Ed.), Progress in Astronautics and Aeronautics: Rarefied Gas Dynamics, vol. 74, AIAA, New York, 1981, pp. 239–255, part I.
[32] G.A. Bird, Perception of numerical method in rarefied gasdynamics, in: E.P. Muntz, D.P. Weaver, D.H. Capbell (Eds.), Rarefied Gas Dynamics: Theoretical and Computational Techniques, in: Progress in Astronautics and Aeronautics, vol. 118, AIAA, New York, 1989, pp. 211–226.
[33] M.A. Gallis, J.R. Torczynski, D. Rader, G.A. Bird, Convergence behavior of a new DSMC algorithm, J. Comput. Phys. 228 (12) (2009) 4532–4548.
[34] M.A. Gallis, T.P. Koehler, J.R. Torczynski, S.J. Plimpton, G. Papadakis, Molecular-level simulations of turbulence and its decay, Phys. Rev. Lett. 118 (6) (2017) 064501.
[35] D.E. Boylan, J.L. Potter, Aerodynamics of typical lifting bodies under conditions simulating very high altitudes, AIAA J. 5 (1967) 226–232.
[36] J.F. Padilla, I.D. Boyd, Assessment of rarefied hypersonic aerodynamics modeling and windtunnel data, AIAA Paper 2006-3390, 2006.